

AI for Computational Pathology and Spatial Omics

From whole-slide images and cell graphs to a survival model and a draft report

Presenter Name, Role

Department, Institution

Meeting name · Month DD, YYYY

Agenda

Pattern: agenda / roadmap

- | | | |
|-----------|-----------------------------------|--|
| 01 | Background | Why tissue architecture matters |
| 02 | Data and methods | Slides, spatial omics and the survival model |
| 03 | Results | Risk score and model comparison |
| 04 | Report-drafting agent | LLM / VLM agent that writes a draft report |
| 05 | Limitations and next steps | What the results can and cannot support |

02

Data and methods

Slides, spatial omics and the survival model

Background

Pattern: big statement

Whole-slide images can be learned from slide-level labels alone, without drawing a single region.

- **Weak supervision:** attention-based multiple-instance learning finds the regions that drive a slide label
- **Scale:** one label per slide makes routine archives usable, not only annotated sets
- **Open question:** the same idea for outcome (survival), and for spatial omics next to H&E

Problem, Approach and Impact

Pattern: problem → approach → impact



PROBLEM

Spatial context is lost

Tile models average away
where cells sit



APPROACH

Learn from the tissue

Foundation-model tiles plus cell
graphs, trained on outcome



IMPACT

Stratify patients by risk

A reproducible score and a draft
report for review

Goal: a risk score from routine slides and spatial data that a pathologist can check.

Data Modalities

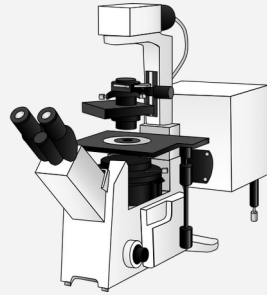
Pattern: data modalities grid



H&E whole-slide images

Scanned at 40×, one per resection

n = XX slides



Multiplex imaging

40-plex protein panel on a subset

n = XX regions



Single-cell phenotypes

Cell types called from the panel

n = XX cells



Clinical and outcome

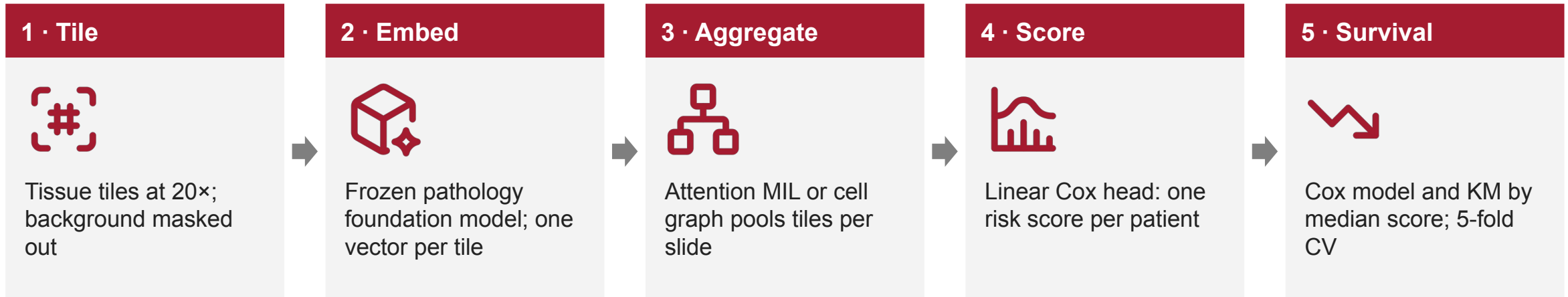
Stage, therapy, recurrence-free survival

n = XX patients

Four modalities from the same patients, linked by patient ID; multiplex data exist for a subset only.

Methods — Slide to Survival

Pattern: methods flow

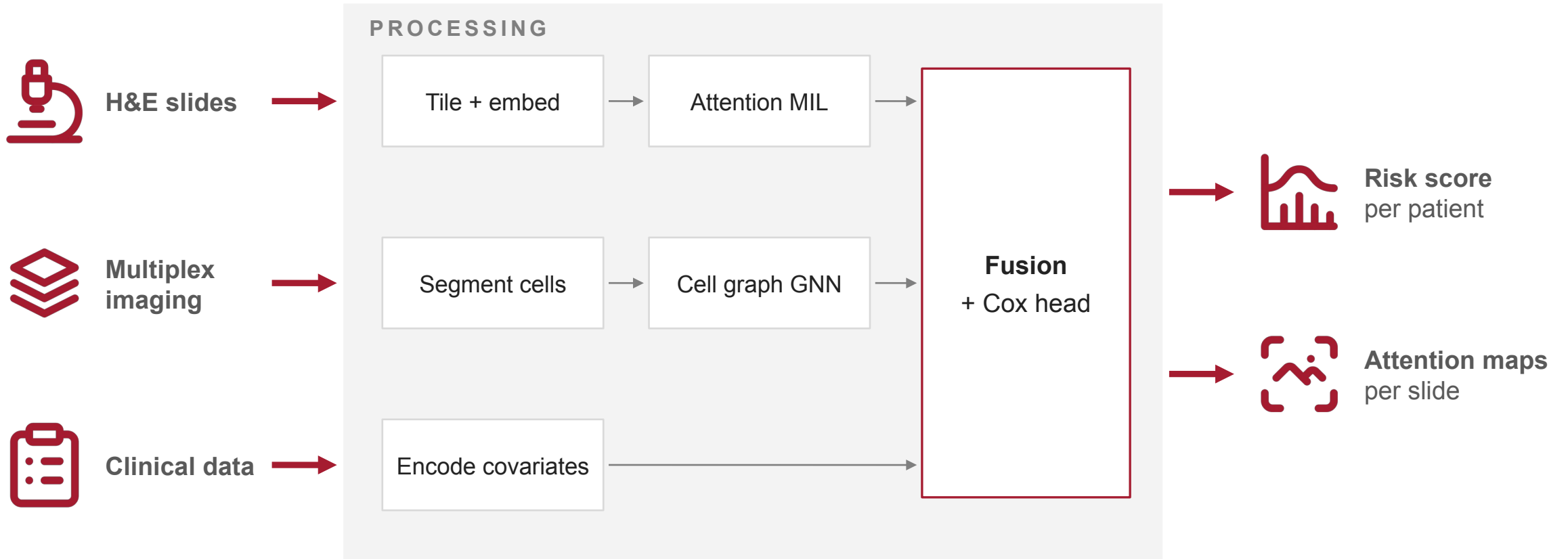


- **MIL** (multiple-instance learning): the slide label is known, the tile labels are not
- **Foundation model**: pretrained on unlabeled tiles; its weights stay frozen, only later layers train
- **Folds split by patient**, so tiles of one patient are never in both training and test

Tiling and embeddings are fixed before outcomes are seen; only the aggregator and the Cox head are trained on survival.

Framework Overview

Pattern: framework diagram



Image, cell-graph and clinical features are fused into one patient-level risk score.

Report-drafting Agent



Tools the agent can call



VLM slide reader

Describes a region the model attended to



Feature store

Returns the patient's risk score and inputs



Guideline lookup

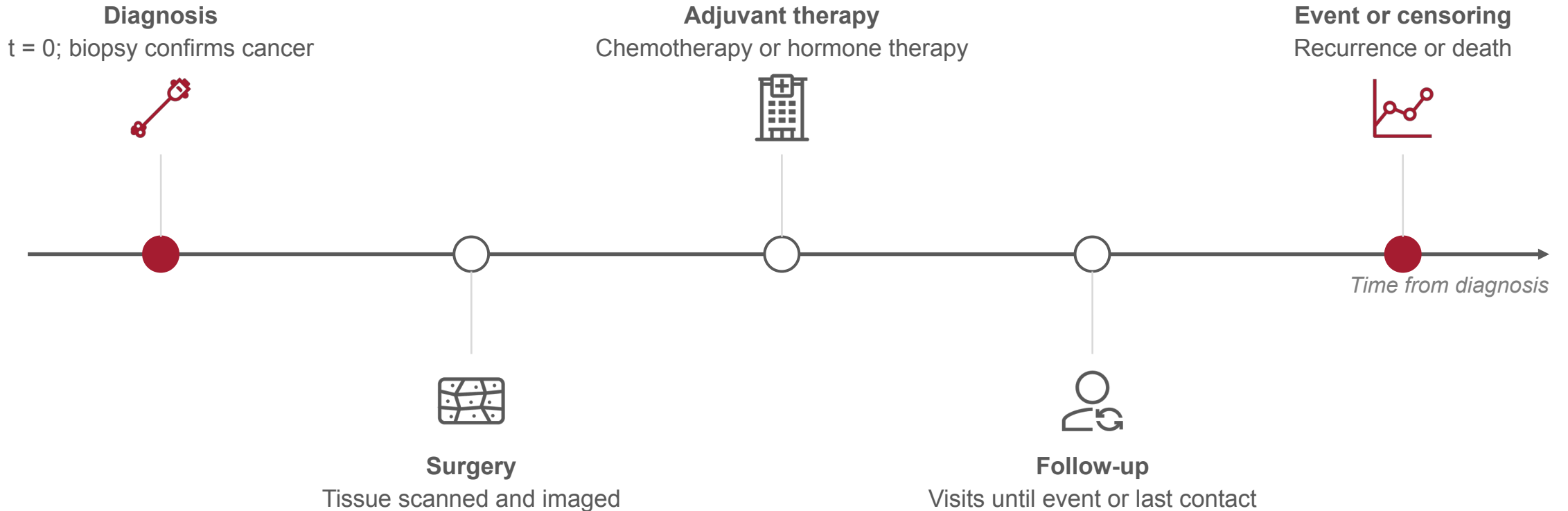
Retrieves the reporting criteria text

Stop: draft goes to a pathologist for sign-off

The agent drafts the report; a pathologist signs it off, and every statement points to a tool output.

Study Design — Patient Timeline

Pattern: timeline



Recurrence-free survival runs from diagnosis to recurrence or death; in GBSG2, 299 of 686 patients had an event and the rest were censored at last follow-up.

Prior Approaches vs This Work

Pattern: two-column comparison

	Prior approaches	This work
Labels needed	⊗ Pixel-level annotation	✓ Slide-level outcome only
Spatial context	⊗ Tile features averaged	✓ Cell graph keeps neighbors
Modalities	⊗ H&E only	✓ H&E, multiplex and clinical
Validation	✓ Often external cohorts	⊖ Internal 5-fold CV so far
Report	⊗ Written by hand	✓ Agent drafts, human signs off

This work trades external validation for richer inputs; the gap is listed under limitations.

Limitations

Pattern: limitations

	LIMITATION	WHAT IT AFFECTS
1	Single-center cohort	Other scanners, stains and sites are untested
2	No external validation yet	Performance is cross-validated on one cohort
3	Multiplex data on a subset (n = XX)	Fusion model trained on fewer patients
4	Agent reports not yet graded	Draft accuracy unknown; drafts only

These results are exploratory; an external cohort is needed before any clinical claim.

Summary

1 Routine slides and spatial data feed one risk score

Tiling and embeddings fixed; only the aggregator and Cox head see outcomes (methods flow)

2 Cell graphs add the spatial context that tile averages lose

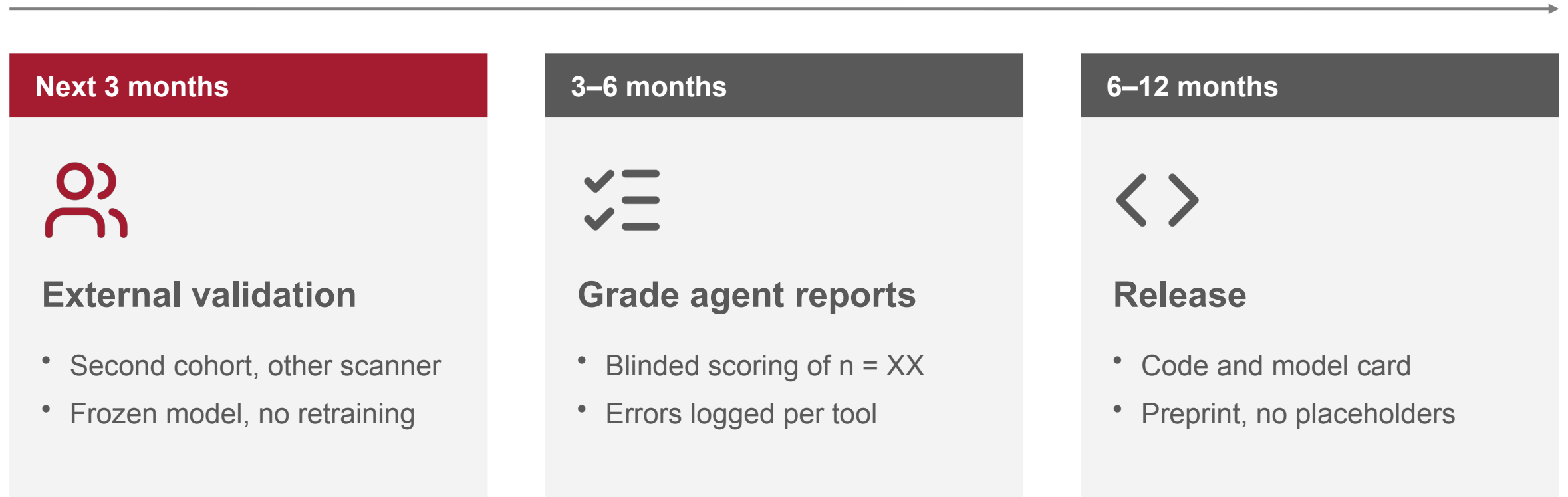
C-index XX vs XX, 5-fold CV (n = XX patients); placeholders for this example

3 An agent drafts the report; a pathologist signs it off

Every statement links back to a tool output (agent loop)

Next Steps

Pattern: next steps / roadmap



External validation comes first; its result decides whether the later steps go ahead.

Thank you

Questions and discussion



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github.com/<org>/<repo>

Supplementary — Model Settings

Pattern: supplementary (hidden)

Component	Setting	Value	Fixed before outcomes?
Tiling	Tile size; magnification	256 × 256 px; 20×	Yes
Tissue mask	Threshold on saturation	Otsu	Yes
Foundation model	Encoder; embedding dimension	Frozen ViT; XX	Yes
Attention MIL	Hidden units; dropout	XX; XX	Yes
Cell graph	Edges; neighbors per cell	k-nearest neighbors; k = XX	Yes
Cox head	Penalty; learning rate; epochs	L2 XX; XX; XX	Tuned in inner folds
Cross-validation	Folds; split unit; random_state	5; patient; 0	Yes
Agent	Model; temperature; max tool calls	XX; 0; XX	Yes

All settings except the Cox penalty were fixed before outcome data were used; random_state is recorded in run_meta.json.

Credits

Source	Icons used in this deck	License
Health Icons (healthicons.org)	biopsy, tissue, hospital, regular-patient, chart-line	CC0 1.0
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	single-cell-umap — James Lloyd; patient — Marcel Tisch	CC0 1.0

Data: GBSG2 trial counts (686 patients, 299 recurrence-or-death events) via lifelines load_gbsg2.

Reference: Lu, Ming Y., et al. *Nature Biomedical Engineering* 5.6 (2021): 555–570.

All other content is an illustrative example; values marked XX are placeholders.